

Lifetime Asset Allocation in New Zealand: A Simulation and Optimization Approach

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1 Foreword and Research Context

This research is conducted in partnership with Consilium NZ LTD (hereafter, Consilium). Consilium is a New Zealand-based financial services company that provides tools and consulting services to financial advisers and their clients in areas including investment management, portfolio construction, risk management, and financial planning. Consilium also offers managed funds and portfolio solutions designed to support advisers in meeting a wide range of client needs. The aim of this research partnership is to examine and improve lifetime asset allocation strategies for individual investors, with particular focus on retirees in the New Zealand context. Our objective is to produce insights and recommendations that may help Consilium better support advisers and their clients.

Although this research is conducted in partnership with Consilium, the views expressed in this paper are ours and do not necessarily reflect the official policy or position of Consilium. We retain full academic independence in the design, analysis, and reporting of this work. We also retain full ownership of the intellectual property associated with this research, while granting Consilium a non-exclusive licence to use the findings internally and share them with clients and partners. No confidential or proprietary information provided by Consilium has been used in this study.

We thank Consilium for their support and collaboration in this research project, and for the feedback that has helped shape the direction and focus of this work.

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2 Introduction

Lifetime asset allocation is one of the highest-impact financial decisions households make.

In general, sources of higher asset returns are associated with higher risks (e.g., (Campbell, 1996; Ghysels et al., 2005; Lundblad, 2007)). This makes the task of selecting an optimal allocation strategy a complex balancing act, weighing the investor’s need for growth to fund future spending against their tolerance for volatility and potential losses. The problem is further complicated by the fact that very few of these factors are known in advance: future returns, future spending needs, and even the investor’s own risk tolerance can all evolve over time in response to changing circumstances.

Put simply, the problem of lifetime asset allocation is one of making the best possible decisions in the face of deep uncertainty about the future. The answer to this problem matters at both household and fiscal levels. The global asset-backed pension market is large (OECD, 2025), and in New Zealand retirement outcomes directly interact with public finances through New Zealand Superannuation. Treasury projections indicate substantial long-run pressure from population ageing (The Treasury, 2025), while policy and industry commentary increasingly emphasises higher private saving and better KiwiSaver decisions (Dexter, 2025; Gilbert, 2025; Luxon, 2025; Palmer, 2025). Better allocation policy therefore has private welfare value and public-finance relevance.

Although many investors spend considerable energy selecting particular securities or funds, Ibbotson and Kaplan (2001) find that roughly 90% of the variation in fund returns is explained by the allocation to major asset classes such as equities, bonds, and cash. As a result, allocation to these broad asset classes is the primary driver of a portfolio’s long-term behaviour. This is even more true for typical retirement portfolios (and especially products like KiwiSaver) which are usually held in managed funds. In New Zealand, KiwiSaver Schemes typically do not allow the investor to exercise granular control over security selection, but rather give them a choice between a small number of pre-defined strategies allocating to different amounts “Growth” and “Income” asset classes (Financial Markets Authority, 2015). So called “Growth” assets typically comprise a mix of domestic and international equities, while “Income” assets are a mix of domestic and international bonds and cash. In this context, the key decision for investors is not which specific stocks or bonds to hold, but rather how much of their portfolio to allocate to Growth versus Income assets, and therefore how much risk to take on.

A well performing allocation strategy is one that maximises the investor’s expected welfare, which is a function both of the expected return and the protection against adverse outcomes.

Where the investor’s needs are fixed, Merton (1969) shows that the optimal risky-asset share is constant through time. There are many simplifying assumptions in this result, but probably the most significant for lifecycle investing is that the investor’s spending needs are fixed and known in advance. Broadly speaking, the investor’s lifecycle can be divided into two phases: accumulation and decumulation. During the accumulation phase, the investor is typically working and saving, building up their wealth over time. When they transition to retirement, they enter the decumulation phase, where they draw down on their accumulated wealth to fund their spending needs.

Bodie et al. (1992) adds in the concept of human capital, which represents the present value of future labor income. When human capital is bond-like, younger households can rationally hold more equity in financial wealth and reduce equity with age (insofar as they continue to be employed). This gives the main theoretical foundation for the intuitive idea that younger investors should hold a higher proportion of their portfolio in risky, high-return assets, and then gradually reduce this proportion as they approach retirement age.

This mirrors the large number of intuitively appealing, easy-to-communicate age-based rules that have become the industry standard for retirement investing. These include the various age-in-bonds rules, 120-minus-age rules, and target-date funds which are prominent in the literature and discourse. Collectively, these types of strategies where asset allocation is a deterministic function of age are often (and hereafter) referred to

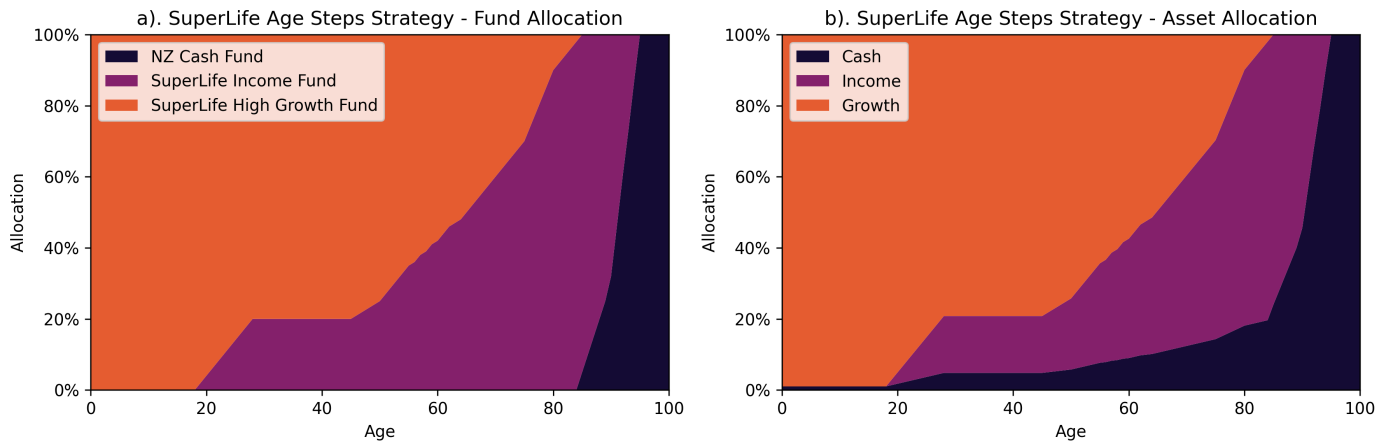


Figure 1: SuperLife's Age Steps target strategy as per their Statement of Investment Policy and Objectives (SIPO), showing the allocation to different funds (a) and the resulting allocation to different asset classes (b). Source: SuperLife (2026).

as “glidepaths”. A great deal of work has been committed to studying the optimal shape of these glidepaths, both in terms of their theoretical properties and their empirical performance. They have been developed into a myriad of financial products and are widely recommended by financial advisers and retirement planning tools.

2.1 The Glidepath Debate: Rising vs Declining vs U-Shaped, Static vs Dynamic

What emerges from the literature is a complex and unresolved debate about the optimal shape of the glidepath, with different studies reaching different conclusions about whether risk should decline, rise, or remain constant over time, and whether the optimal strategy differs between the accumulation and decumulation phases.

There are essentially five main shapes that have been proposed for the optimal glidepath (see Figure ??):

- Declining equity (DE): the conventional wisdom that risk should decline with age, often implemented as a simple age-based rule.
- Rising equity (RE): the contrarian view that risk should actually increase with age, particularly at the start of retirement, to capture the higher expected returns of equities when they are most needed.
- U-shaped: declining equity during the accumulation phase, followed by rising equity during retirement.
- Inverted U-shaped: rising equity during the accumulation phase, peaking at retirement, followed by declining equity during retirement.
- Static allocation: the view that a constant equity share throughout life is optimal, and that the shape of the glidepath is less important than the overall level of risk taken.

The reasons for this disagreement are complex, but a key fault line in the literature is the choice of methodology for generating return scenarios and evaluating strategies. Different studies use different approaches, including historical rolling periods, Monte Carlo simulation with IID returns, and block bootstrap methods that preserve time-series dynamics. These methodological choices have a significant impact on the results and conclusions drawn about the optimal glidepath shape.

The standard industry recommendation is that investors decrease their risk exposure as they age, continuing to reduce risk even after retirement. This approach appears in investment textbooks (e.g Bengen, 1996; Bodie et al., 2023) and is widely recommended by financial advisers and retirement planning tools (Choi, 2022). A review of U.S target date funds (TDFs) by Dolvin et al. (2010) find that most follow some form of declining-risk structure, often resembling a 120-minus-age equity rule. For a domestic example, take the SuperLife Age Steps allocation profile shown in Figure ??. At one point a declining-risk glidepath was considered as part of potential reforms to the United States Social Security system under the George W. Bush administration (see R. Shiller, 2005; R. J. Shiller, 2006).

However, declining-risk glidepaths have been challenged by a substantial body of academic work, and the post-retirement period in particular is a site of active and unresolved disagreement in the literature. Responding

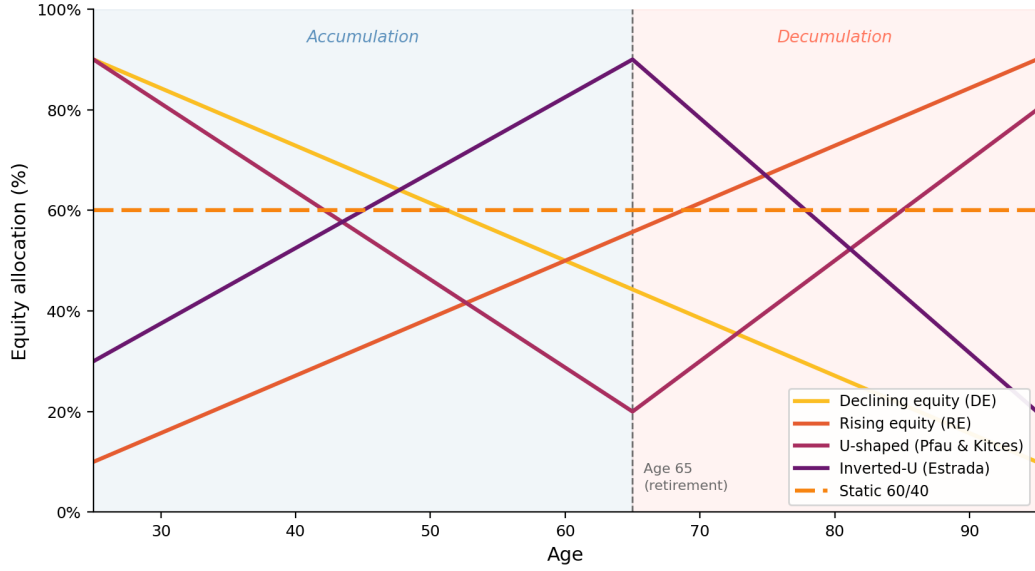


Figure 2: Illustration of different proposed glidepath shapes. The x-axis represents age, with the vertical dashed line indicating the point of retirement. The y-axis represents the equity share of the portfolio. The different lines represent the different proposed glidepath shapes: declining equity (DE), rising equity (RE), U-shaped, inverted U-shaped, and static allocation.

to the aforementioned Bush-era proposal, R. Shiller (2005) and R. J. Shiller (2006) find that declining-risk glidepaths often fail to outperform a 100% equity strategy. Estrada (2014) and Estrada (2015) find that declining-risk strategies tend to underperform constant-risk alternatives or even contrarian rising-risk strategies across a wide range of countries and historical periods.

Studies generating return scenarios using Monte Carlo simulation with IID returns tend to find support for U-shaped glidepaths. This means a declining-risk strategy during the accumulation phase, equity exposure low around the point of retirement, and then rising equity during the retirement phase. The rationale is that starting conservatively limits the damage from a bad early period (which is particularly harmful at the point of retirement when the portfolio is largest), while systematically accumulating equities at depressed prices through a downturn allows the portfolio to capture the eventual recovery when growth is most needed. The proposed risk mechanism that this avoids is a sequence-of-returns argument, that is the risk of a series of bad returns early in retirement that depletes the portfolio when it is largest and most vulnerable. In this line of thinking, the accumulation phase is more supportive of a DE strategy, because the sequence risk mechanism is less relevant when the portfolio is still growing and not yet being drawn down. Here the investor can afford to be more aggressive early on, capturing the higher expected returns of equities when they have the longest time to compound, and then gradually reduce risk as they approach retirement and peak wealth. This is the conventional lifecycle logic that underpins most age-based glidepaths. Proponents of a U-shaped glidepath argue that the same logic applies in reverse during retirement: starting conservatively at the point of retirement limits the damage from a bad early period, while systematically accumulating equities at depressed prices through a downturn allows the portfolio to capture the eventual recovery when growth is most needed.

Pfau and Kitces (2014) make the case for rising equity during retirement using Monte Carlo (MC) simulation. Across 10,000 simulated retirements with lognormal IID returns, 121 glide path combinations (varying start and end equity from 0% to 100%), and both 4% and 5% withdrawal rates, they find that rising equity during retirement reduces both the probability and magnitude of portfolio failure. The optimal starting allocation is 20–40% equity, rising to 60–80% by end of retirement. A 30%→60% rising path beats a static 60/40 on failure rate *and* failure magnitude. This holds across three capital market assumption sets. They argue that retirement outcomes are almost entirely determined by the first 15 years. A RE strategy during retirement starts conservative, limiting the damage from a bad early period (and at peak asset accumulation), and then systematically accumulates equities at depressed prices through the downturn.

Anarkulova et al. (2025) use a block-bootstrap methodology applied to a dataset of monthly real returns across 39 developed countries from 1890 to 2023, drawing on over 31,800 country-months of data and preserving the time-series and cross-sectional structure of historical returns. Under block bootstrap, the optimal allocation is approximately 100% static equity throughout the lifecycle. The sole exception is a brief tactical allocation to bills at the point of retirement, which does resemble a kind of aggressive U-shaped glidepath, but they argue that this is a special case, driven by the rigidity of a fixed real 4% withdrawal rule (based on Bengen, 1994)

and when spending flexibility is restored, this exception disappears and the all-equity result holds at every age. They also argue that the conventional preference for bonds is substantially overstated by return-generation decisions in the literature. They compare optimal glidepaths under IID return sampling and block bootstrap, and find that the optimal bond allocation is higher under IID assumptions. They argue that this is because the long-horizon properties of bonds are much less favourable than their short-horizon statistics suggest. At 30-year horizons, bond variance *increases* relative to the one-year level (variance ratio 2.30), whereas equity variance *decreases* (variance ratio 0.75). Bonds also carry a correlation of -0.78 against inflation, which they argue destroys real purchasing power in precisely the environments most harmful to retirees.

In essence, Anarkulova et al. (2025) argue that IID assumptions overstate the risks of sequence-of-return risk. In reality, timeseries effects mean the sequence is actually autocorrelated and mean-reverting. In this environment, the higher expected returns of equities are more reliably captured over long horizons and the risks of bonds are more pronounced.

Using deep reinforcement learning to solve a comprehensive lifecycle model against Monte Carlo simulated returns (with autocorrelated return dynamics), Duarte et al. (2021) find the optimal equity allocation at retirement is approximately 60%, and that TDFs (which land at 30–40% equity) are too conservative; the welfare cost of following a TDF rather than the optimal policy is 1.7–2.8% of lifetime consumption. However, they otherwise find that a declining trajectory of equity allocation proposed by many TDFs (declining equity during accumulation) is broadly optimal, and that the optimal trajectory post-retirement is approximately flat (not declining or rising).

However, there is considerable scepticism toward conventional declining-equity strategies during both the accumulation and decumulation phases. The argument is essentially that a DE strategy lowers the expected return of the portfolio precisely when the portfolio is at its largest: lifecycle funds are aggressive when the portfolio is small (early career) and conservative when it is large (near retirement), which is backwards. Estrada (2014) establishes this for the accumulation period. The companion paper, Estrada (2015), applies the same DMS dataset to the retirement drawdown period, testing 81 rolling 30-year retirement windows starting from 1900 with a 4% real withdrawal rule. The key finding reverses the accumulation result: DE strategies have lower failure rates *and* higher terminal wealth than rising-equity mirrors in every specification. In the United States, DE failure rates range from 4.9 to 8.6%, while the corresponding RE mirrors fail at rates of 8.6 to 21.0%. Across all 19 countries, DE fails at 31–32% versus RE at 39–49%. The mechanism is a sequence-of-returns argument specific to the drawdown context. A retiree’s portfolio is largest at the point of retirement; DE starts equity-heavy at this point, capturing equity gains while the portfolio is large and compounding on a large base. As the portfolio shrinks through withdrawals, the gradual shift to bonds provides capital preservation. RE strategies invert this: equity exposure builds only after withdrawals have already depleted the portfolio, so the eventual equity gains arrive too late and on too small a base to matter. Combined with the 2014 accumulation findings, Estrada’s recommended full-lifecycle shape is an *inverted U*: equity should increase during working life, peak at the point of retirement, and then decline through the drawdown period.

The DE position is further weakened by evidence that even dynamic DE management adds little or no value over a well-chosen static allocation. Blanchett (2007) tests 43 glide paths (11 static allocations and 4 types of declining-equity dynamics (linear, stair-step, concave, and convex)) bootstrapped from 80 years of monthly return data, across 1,071 scenarios defined by distribution period (20–40 years) and real withdrawal rate (3–8%). Rising-equity paths are not tested; the comparison is strictly static versus DE. Static allocations have the lowest failure probability in all but one of the 1,071 scenarios. The best static portfolio by raw failure probability is 100% equity; using his “Success to Variability Ratio” (probability of success divided by portfolio standard deviation), the best risk-adjusted portfolio is a 60/40 stocks-bonds allocation.

An important contextual observation from Blanchett (2007) is that the choice of glidepath matters little when withdrawal rates are moderate and horizons are short. At a 4% withdrawal rate over a 20-year period, the range of failure probabilities across all 43 strategies is only 1.56 percentage points. The same metric reaches 62.45 percentage points at a 6% withdrawal rate over 40 years. The glidepath debate is therefore most consequential for aggressive scenarios, and the literature’s disagreements should be interpreted in that light.

The tension between these results is not easily resolved, but it is clear that the choice of return-generation methodology is a key driver. Additionally, the choice of risk measure (failure probability, expected shortfall, risk-adjusted success ratio, expected utility) and the specification of the spending rule (fixed real withdrawal, flexible spending, etc.) also matter. Pfau and Kitces (2014) use Monte Carlo with independently and identically distributed (IID) lognormal returns, in which bad returns are equally likely at any point and carry no autocorrelation.

Historical rolling periods (Estrada, Bengen) preserve the actual structure of return sequences. They argue that because recessions cluster, valuations mean-revert, and bad periods tend to be followed by recoveries,

IID assumptions overstate the risks of sequence-of-returns risk as stocks are more likely to recover from an early dip than the IID model suggests. This approach does, however, limit the sample size to the number of non-overlapping historical periods, which is a particular problem for long-horizon retirement planning where the number of 30- or 40-year periods in the historical record is very small. Anarkulova et al. (2025) use a block bootstrap approach, in which 6–12 month blocks of returns are sampled from the historical record and concatenated to create synthetic return paths. This preserves some of the time-series dynamics of returns, including mean reversion and volatility clustering, while generating a much larger sample of data. However, even in this approach it is unclear how “novel” return paths are, particularly given the small number of downturns even in a long historical record. This means that the optimisation could be overfitting to a small number of historical downturns with the particular time-series dynamics that they exhibit. Duarte et al. (2021) use a Monte Carlo approach with autocorrelated return dynamics, which might arguably be more realistic than IID assumptions, but still relies on a parametric model of return dynamics that may not capture all relevant features of the data. Interestingly, they find some support for the typical DE trajectory during the accumulation phase (typically recommended by TDFs), but not during the retirement phase, where they find a flat trajectory is optimal.

The choice of objective function also matters. Pfau and Kitces specifically target failure risk minimisation; Estrada targets both failure risk and terminal wealth jointly and finds DE dominates on both. Blanchett introduces the risk-adjusted Success to Variability ratio and finds it tilts the optimal allocation toward moderate equity, regardless of glidepath shape. Studies optimising expected utility with explicit risk aversion parameters (e.g. Duarte et al. (2021)) tend to find approximately constant, high equity allocations consistent with neither the U-shape nor the inverted-U, but broadly aligned with Blanchett’s static recommendation.

Finally, the specification of the investor’s spending and income needs is a critical driver of optimal policy. In discussing the paper by Anarkulova et al. (2025), Felix (2025) argues that the real risk to investors is not the risk of portfolio failure per se, but the risk of being unable to adjust spending in response to market conditions. For our purposes, investor spending should be grounded in realistic benchmarks of actual spending patterns in retirement, which are not necessarily well captured by a fixed real withdrawal rule.

2.2 Problem Statement and Research Questions

The review above reveals that the literature has not converged on a single answer to even the most basic question of optimal retirement allocation, and that disagreements are substantially driven by modelling choices rather than fundamental differences in the underlying economic problem. The choice of return-generation methodology (IID Monte Carlo, historical rolling periods, or block bootstrap), the choice of risk or objective measure (failure probability, terminal wealth, expected utility), and the specification of the spending rule (fixed real withdrawal vs. flexible spending) each independently shift the apparent optimal policy. A secondary but practically important question is whether the glidepath shape matters at all compared with the more fundamental question of whether bonds should be held in retirement portfolios at the levels currently recommended.

This thesis addresses the following core problems in the New Zealand context: *(i) What lifetime allocation policy is most appropriate for New Zealand retirees under realistic return dynamics and spending paths? (ii) How sensitive are these recommendations to the modelling choices that drive disagreement in the literature? (iii) Can allowing allocation policy to respond to both age and wealth improve outcomes beyond what age-only policies achieve?*

Concretely, we ask:

1. Are conventional age-based declining-risk glidepaths optimal under standard assumptions, and does that result survive under NZ-calibrated return dynamics and spending paths that decline in real terms with age?
2. How sensitive are “optimal” recommendations to modelling assumptions—methodological (return-generation process, IID vs. block bootstrap, risk measure, spending rule) and substantive (household composition, housing tenure, real spending trajectory, longevity horizon, NZ Super as a guaranteed income floor)?
3. Does allowing allocation policy to depend on both age and wealth improve outcomes relative to age-only glidepaths, and if so, under what conditions?

2.3 Our Approach and Specification

We make three contributions. First, we synthesise the lifecycle allocation literature with explicit attention to where theory, empirical evidence, and industry implementation diverge—and where that divergence is at-

tributable to modelling choices rather than genuine empirical disagreement. Second, we calibrate retirement spending to New Zealand institutions: we use empirically grounded spending trajectories that decline in real terms after retirement rather than a fixed real withdrawal rule, and we treat NZ Super as a guaranteed income floor that materially reduces the required drawdown from financial assets. Third, building on Mäkinen and Toivanen (2024), we implement a flexible Monte Carlo optimisation framework that can evaluate both age-based glidepaths and wealth-responsive policies across alternative return processes and risk measures.

The framework is intentionally modular: allocation policy class, spending rule, risk objective, and return model can each be varied independently. This is necessary because, as the preceding review makes clear, the disagreement in the literature is not about a single parameter but about the structure of the model itself. Our goal is therefore not to produce one universal glidepath recommendation; it is to provide a defensible, testable method for identifying which modelling choices drive conclusions, and to characterise the policy-relevant strategies that are robust across plausible specifications in the New Zealand setting.

3 Retirement in New Zealand

The New Zealand-specific literature on retirement concerns remains limited, so we draw extensively on international evidence. Understanding the institutional context is essential for interpreting this evidence appropriately. New Zealand’s retirement system combines a universal public pension (New Zealand Superannuation) with voluntary private savings through KiwiSaver. New Zealand Superannuation provides all residents aged 65 and over with a baseline income regardless of work history or assets, and adjusts annually for inflation. KiwiSaver allows voluntary contributions from employees, matched by employers and the government. Together, these mechanisms provide a safety net while encouraging personal saving (Ministry of Social Development, 2022; Te Ara Ahunga Ora: Retirement Commission, 2021).

KiwiSaver was introduced in 2007, so most current retirees have little or no accumulated balance. Currently, 40 percent of New Zealanders aged 65 rely entirely on New Zealand Superannuation, with a further 20 percent holding minimal additional savings (Retirement Commission Te Ara Ahunga Ora, 2022). As the scheme matures and government policy encourages higher contribution rates (Luxon, 2025), future cohorts will retire with substantially larger balances. This demographic shift increases the need for robust guidance on asset allocation and withdrawal strategies during retirement.

Given these trends, the Australian experience becomes increasingly relevant. Australia’s retirement system combines a means-tested public pension with mandatory superannuation contributions of 12 percent of income (Australian Taxation Office, n.d.; Swoboda, 2014). While institutional differences remain, the growing role of KiwiSaver makes the two systems more comparable than in the past.

3.1 New Zealand Superannuation Policy Settings

New Zealand Superannuation (NZ Super) is a universal, non-means-tested public pension paid from age 65, funded from general taxation rather than a contributory insurance scheme. It forms the guaranteed income floor underpinning all retirement planning in New Zealand. Entitlement is not conditional on prior employment, investment, or KiwiSaver participation; a recipient’s private wealth or income has no effect on the payment rate.

Eligibility

The primary eligibility criteria are age and residency. A person becomes entitled upon turning 65, provided they have been both resident and present in New Zealand for a sufficient period since age 20. For those born on or before 30 June 1959, the requirement is 10 years total, of which 5 must have accrued since age 50. For those born on or after 1 July 1977, the total requirement is 20 years (with 5 since age 50), reflecting a graduated increase introduced by the Social Assistance (Residency Qualification) Legislation Act 2018. Special provisions apply to recognised refugees and to periods spent in the Cook Islands, Niue, and Tokelau (New Zealand Parliament, 2001, s. 8).

Payment Rates

Rates are set in Schedule 1 of the New Zealand Superannuation and Retirement Income Act 2001 and adjusted each 1 April by Order in Council. The April 2026 rates (before deduction of tax under code M) are listed in Table 1.

Table 1: NZ Super rates from 1 April 2026 (before tax). Source: Work and Income New Zealand (2026a, 2026b).

Rate type	Per week	Per fortnight	Per year
Single living alone	\$647.37	\$1,294.74	\$33,663
Single sharing	\$595.57	\$1,191.14	\$30,970
Couple (per person)	\$492.14	\$984.28	\$25,591
Couple (combined)	\$984.28	\$1,968.56	\$51,182

After deduction of standard tax (code M), the fortnightly after-tax rates are approximately \$1,110 for a single person living alone and \$1,708 combined for a couple, equivalent to roughly \$28,900 and \$44,400 per year respectively (Work and Income New Zealand, 2026a, 2026b). These figures differ slightly from those used in Matthews (2025) (who report \$519 and \$799 per week for singles and couples respectively) due to differing tax rate assumptions; this study uses the WINZ statutory rates as the primary reference.

Indexation and the Wage Floor

The Act mandates annual adjustment for CPI inflation (so the rate cannot decline in real terms) subject to a binding wage floor. The combined after-tax couple rate must remain between 66 and 72.5 percent of net average ordinary-time weekly earnings, as measured by the Quarterly Employment Survey (New Zealand Parliament, 2001, s. 16). In practice, wages have grown faster than CPI in recent decades, meaning the wage floor has been the binding constraint. The single living alone rate is fixed by statute at 65 percent of the couple combined rate (after tax), and the single sharing rate at 60 percent. This legislated floor is a key feature distinguishing NZ Super from defined-benefit pension schemes that index only to prices.

Policy Adjustments and Reform Debate

Despite ongoing fiscal pressure from population ageing, NZ Super’s core parameters have remained largely unchanged since 2001. The most significant recent amendment was the gradual increase in the residency qualifying period from 10 to 20 years for those born after 1 July 1977, phased in from 2018. The eligibility age of 65 has not changed; a proposal to raise it to 67 was advanced and subsequently abandoned. A 2022 government review (the Retirement Review Issues Paper, conducted by Te Ara Ahunga Ora) examined a range of possible adjustments including age, means-testing, and residency, without proposing binding changes (Ministry of Social Development, 2022).

Supplementary Benefits

Several additional transfers are available to eligible retirees:

- **Winter Energy Payment:** a universal supplement paid automatically to NZ Super recipients from 1 May to 1 October each year. The 2026 rate is \$20.46 per week for single people and \$31.82 per week for couples and people with dependants (Work and Income New Zealand, 2026a). No application or means test is required.
- **Accommodation Supplement:** a means- and asset-tested housing cost subsidy available to renters, boarders, and mortgage holders across all age groups including retirees. The amount varies by accommodation costs and location zone, with higher rates in Auckland and Wellington. Eligible retirees with moderate housing costs may receive up to several hundred dollars per fortnight (Work and Income New Zealand, 2026a).
- **SuperGold Card:** a concession card issued automatically to NZ Super recipients, providing discounts on public transport (including free off-peak travel on most regional services), selected retail, and other services (Ministry of Social Development, 2022).
- **Residential Care Subsidy:** an asset-tested subsidy for long-term residential aged care (rest homes and dementia units). As at 2022, a single person is generally required to contribute their own assets up to a threshold of around \$240,000 before the subsidy takes effect, with the family home exempt for a defined period if a spouse, dependent, or carer remains in residence (Ministry of Social Development, 2022).

Together, NZ Super plus the Winter Energy Payment provide a reliable inflation-linked income floor for all retirees.

3.2 Patterns of Spending in Retirees - New Zealand and Abroad

3.2.1 Spending By Age in New Zealand

The Stats NZ Household Economic Survey (HES) provides the most comprehensive data on New Zealand retirement spending patterns. Two recent reports are particularly relevant. Le and Richardson (2023) analyse expenditure patterns by age across New Zealand households, while Retirement Income Interest Group (2024) focus specifically on spending through retirement and its implications for drawdown strategies.

Using HES data for the year ended June 2019, Le and Richardson (2023) find that expenditure declines sharply at retirement and continues to fall thereafter. Mean household expenditure is \$64,700 for ages 65–74, then 28% lower for ages 75–84, and 44% lower for ages 85+. Expenditure declines across most categories, with the largest falls in mortgage repayments, air travel, and recreation. Unusually, healthcare expenditure also declines with age in New Zealand, likely reflecting the public funding of most core medical services. Retirement Income Interest Group (2024) estimate the real decline at roughly 2% per year after age 65.

The HES data cannot, however, distinguish voluntary from involuntary spending reductions. Retirees may reduce expenditure because they become less active and have genuinely lower needs, or because they face binding financial constraints. Without longitudinal data tracking individuals over time, these explanations cannot be separated cleanly. The observed decline likely reflects both channels.

Understanding the source of this decline is central to evaluating welfare. If spending falls primarily due to financial constraints, optimal strategies should aim to sustain pre-retirement consumption levels. If it reflects voluntary lifestyle adjustment, forcing consumption to remain high may be neither necessary nor welfare-improving. This distinction also affects risk tolerance: financially constrained retirees may need safer portfolios, while those adjusting voluntarily may tolerate more risk.

The retirement spending decline is sometimes called the “retirement consumption puzzle” (Ando & Modigliani, 1963). Standard life-cycle models predict that households should smooth consumption over time: accumulating wealth during working life, then gradually decumulating in retirement while maintaining living standards. Under this framework, retirees should save just enough to sustain pre-retirement consumption, ultimately exhausting wealth near the end of life. Yet empirical evidence shows that many retirees reduce spending at retirement despite having sufficient wealth, and in some cases continue to accumulate wealth during retirement. This suggests either that retirees fail to optimise consumption intertemporally, or that additional factors shape their behaviour.

Whether this pattern reflects genuine welfare loss or merely measurement and composition effects remains unclear. Consumption expenditure is an imperfect proxy for welfare, particularly when higher spending is driven by medical needs or lower spending reflects voluntary lifestyle adjustment. Barrett and Brzozowski (2010) find that households retiring unexpectedly due to adverse health shocks experience sharper spending declines than those retiring as planned, consistent with involuntary constraints. However, much of the observed decline may be voluntary. Retirees spend less on commuting, work clothing, and purchased meals (Aguiar & Hurst, 2008; Battistin et al., 2009), and may substitute market purchases with home production, reducing measured expenditure while maintaining consumption (Becker, 1965; Hurd & Rohwedder, 2003). These mechanisms can mask substantial cross-household heterogeneity in retirement outcomes.

Cross-country comparisons highlight the role of institutional context. Banks et al. (2016) find that U.K. retirees reduce real consumption by 2.2% per year versus 1.4% per year in the U.S. This divergence is driven largely by healthcare financing: U.S. retirees face high out-of-pocket costs that rise with age, while U.K. retirees benefit from comprehensive public coverage. When healthcare spending is excluded, consumption trajectories are more similar across countries.

Australian evidence also shows spending declines of 0.5–1% per year in real terms (Grattan Institute, 2019). Unlike New Zealand’s cross-sectional HES data, the Grattan Institute analysis uses longitudinal data tracking individuals over time. The decline persists in both cross-sectional and longitudinal samples, suggesting it reflects genuine behavioural change rather than cohort effects. Importantly, the Grattan Institute finds that most Australian retirees are not financially constrained: fewer than half draw down wealth at all, and over 40% accumulate wealth during retirement. Retirees also report feeling more financially secure than working-age households. As Retirement Income Interest Group (2024) note, this indicates that spending reductions primarily reflect voluntary adjustment to changing lifestyle needs rather than binding constraints.

3.2.2 Bequests and Precautionary Savings

An aspect of the “retirement consumption puzzle” that has also been studied is why many retirees do not draw down significantly on their wealth during retirement, and in some cases even accumulate more wealth during retirement. This is a well-established phenomenon, having been observed in multiple countries and datasets.

The limited wealth drawdown observed in many countries requires explanation. Hurd (1987) proposed that retirees preserve wealth to leave bequests. However, survey evidence casts doubt on this explanation. Alonso-García et al. (2022) find that Australian and Dutch retirees rarely cite intentional bequest motives beyond provision for a surviving spouse. Instead, retirees prioritise financial independence and quality of life, suggesting that limited decumulation reflects precautionary saving for uncertain future costs.

Precautionary motives are well-supported empirically. De Nardi et al. (2010) show that exposure to uncertain health and long-term care costs explains much of the observed non-decumulation in the United States. Similarly, Banks et al. (2016) find that healthcare costs drive higher late-life spending among U.S. retirees relative to those in the U.K. with broader public coverage. New Zealand’s system is mixed: most core medical services are publicly funded, but long-term care such as assisted living and memory care can involve substantial out-of-pocket costs (New Zealand Government, n.d.). Retirement Income Interest Group (2024) conclude that precautionary motives in New Zealand are likely weaker than in the United States but stronger than in countries such as France and Germany with more comprehensive public provision, placing New Zealand alongside the United Kingdom in a mixed-funding model. In aggregate, the evidence does not support widespread precautionary withholding of savings among New Zealand retirees; the observed pattern of limited drawdown is more likely to reflect a combination of voluntary lifestyle adjustment, genuine uncertainty about care costs, and the buffering role of NZ Super as a guaranteed income floor.

3.2.3 Spending by Household Composition and Wealth

Household composition and housing tenure are among the strongest predictors of retirement spending levels and subjective wellbeing in New Zealand. Le and Richardson (2023) find that single-person retired households spend around \$30,700 per year on average, compared with \$65,100 for couples. Notably, spending increases more than proportionately with household size: the average couple spends more than double the amount of a single-person household, implying weaker economies of scale (in New Zealand at least) than other studies, such as (Anarkulova et al., 2025), have suggested in other contexts. Le and Richardson (2023) attribute the disproportionate gap primarily to differences in discretionary spending: couples devote substantially more to transport and recreation and culture, the two largest discretionary categories. Single-person households living alone are also more likely to be renters or to carry a mortgage, further constraining their discretionary budget, while couples are more likely to own outright.

Homeownership status has a particularly pronounced effect on both spending and wellbeing. Among retired New Zealand households, approximately 66 percent own their home outright, 14 percent carry a mortgage, and 14 percent rent (Le & Richardson, 2023). Outright owners report the highest subjective wellbeing of any tenure group and spend around \$55,800 per year, of which housing costs form a relatively small share. Renters spend considerably less in total, at around \$41,400 per year, but pay approximately \$13,100 of this in rent annually, leaving a significantly lower discretionary budget. Renters also report the lowest wellbeing among all tenure types, even after controlling for income. This suggests that the financial and psychological burden of ongoing housing costs in retirement is substantial and is not fully compensated by lower total spending. This well-being gap will have interesting consequences for asset allocation decisions, especially as it pertains to whole-life allocation and spending strategies. One of the peculiarities of the New Zealand KiwiSaver system is that it allows for the use of accumulated savings to be withdrawn for the purchase of a first home. Balancing the competing goals of homeownership and retirement savings is a unique challenge for New Zealand investors, and one that may have important implications for their lifetime asset allocation decisions. For instance, if utility gained from homeownership is sufficiently high, it may justify depleting the investors retirement savings while young to achieve homeownership, even if this results in a lower savings balance at retirement. Inversely, if buying a house depletes the investor’s retirement savings to a degree that the excess burden of housing costs would have otherwise have been more than offset by the additional savings balance at retirement, it may be optimal to delay homeownership and instead prioritise retirement savings.

The importance of homeownership is expected to increase over time. Retirement Income Interest Group (2024) note that home ownership among New Zealanders aged 65 and over is projected to fall from around 80 percent today to approximately 60 percent by 2048, as younger cohorts face higher barriers to entry in the housing market. This shift implies that expenditure benchmarks calibrated on the current cohort of retirees

(who benefit from unusually high ownership rates) will understate spending requirements for future generations, particularly with respect to housing costs.

3.2.4 Expenditure Benchmarks for New Zealand Retirees

The most widely used empirical benchmarks for New Zealand retirement spending are the annual Retirement Expenditure Guidelines published by the NZ Fin-Ed Centre at Massey University (Matthews, 2025). These are derived from the Statistics New Zealand Household Economic Survey (HES) and reported at two tiers: a “No Frills” budget based on the second income quintile of retired HES households, reflecting a basic standard of living with few luxuries, and a “Choices” budget based on the fourth quintile, reflecting a more comfortable lifestyle with some discretionary spending. The 2025 figures, adjusted to 30 June 2025, are shown in Table 2.

Table 2: New Zealand Retirement Expenditure Guidelines, 2025. Weekly spending, with after-tax NZ Super rates for comparison. Source: Matthews (2025).

Budget	One-person household		Two-person household	
	Metro	Provincial	Metro	Provincial
No Frills (\$/week)	705	581	937	1,061
Choices (\$/week)	791	772	1,780	1,243
NZ Super after tax (\$/week)	519 (single)		799 (couple)	

For the No Frills provincial single-person household, NZ Super alone covers approximately 89 percent of spending, implying a required annual drawdown of only around \$3,200 from private savings. For the No Frills metro single, the gap is somewhat larger at around \$9,700 per year. Couples at the Choices tier, particularly in metro areas, face substantially higher gaps; the Choices metro couple budget of approximately \$92,600 per year implies a drawdown requirement of around \$51,000 annually above NZ Super. However, this upper estimate should be interpreted cautiously, since the fourth income quintile of HES retired households likely includes a material proportion of households still receiving employment or investment income, rather than pure retirees drawing down accumulated savings.

A central limitation of these guidelines is that they are constructed from cross-sectional data and therefore implicitly treat retirement spending as fixed in real terms throughout retirement. The HES samples households of all retirement ages at a single point in time, which means the guidelines effectively average the higher spending of the more numerous newly retired with the lower spending of older cohorts, without capturing the within-person spending trajectory over time (Retirement Income Interest Group, 2024). If real spending declines by approximately 2 percent per year after age 65 (as both Retirement Income Interest Group (2024) and Le and Richardson (2023) find) then the guidelines will overstate the savings balance required at age 65 for any given income target.

Retirement Income Interest Group (2024) quantify this directly. Using a replacement-rate methodology, they calculate that reaching a total income target of \$56,000 per year in real terms for a single retiree, on top of NZ Super, over a retirement from age 65 to 90, requires a savings balance of \$605,000 under a constant-real-spending assumption. If instead nominal income is assumed to remain flat (equivalent to a 2 percent real decline when inflation is also 2 percent), the required balance falls to \$375,000, a reduction of 40 percent. An intermediate assumption of a 1 percent real annual decline yields a balance of \$480,000. These calculations assume a 3.5 percent net investment return.

For the purposes of this study, we treat the expenditure guidelines as an empirical anchor for age-65 spending levels. We calibrate initial retirement spending to the No Frills and Choices benchmarks as the lower and upper bounds for a representative retiree, and model real consumption as declining at approximately 2 percent per year thereafter. NZ Super is treated as a guaranteed, inflation-linked income floor from age 65. Under this calibration, the required drawdown from financial assets diminishes progressively as retirement spending falls toward, and in some cases below, the level already covered by NZ Super. This pattern has material implications for both the required savings balance at retirement and the optimal asset allocation strategy during decumulation.

4 Methods for Simulating Lifetime Wealth Trajectories

A key component of studying lifetime asset allocation is the ability to simulate realistic wealth trajectories for individual investors. This requires modelling both accumulation and decumulation, while accounting for contributions, withdrawals, returns, and spending needs. Our approach uses a Monte Carlo framework that generates a wide range of possible outcomes under alternative allocation policies and market conditions. Importantly, the simulation framework is agnostic to return source: we can use historical, bootstrapped, or model-simulated returns. This flexibility allows us to test strategy robustness across a broad set of assumptions. The framework also accommodates time-varying deposits and withdrawals, which is essential for representing real-world household behaviour.

Unlike the approach proposed by Merton (1969), ours is a discrete-time simulation that models the investor’s wealth at regular intervals. This allows us to capture savings and spending behaviour in fixed-dollar amounts, which is more realistic for most retirees.

Each step of the model takes the following form:

$$W_{t+1} = g_{t+1}W_t + \pi_{t+1} \quad (1)$$

Where W_t is the wealth at time t , g_{t+1} is the growth factor on the portfolio between time t and $t + 1$, and π_{t+1} is the net contribution (or withdrawal if negative) made at time $t + 1$. This can represent regular contributions during the accumulation phase or withdrawals to cover spending needs during the decumulation phase. The model can be easily adapted to include varying contribution and withdrawal amounts over time, allowing for a more nuanced representation of investor behaviour, including simulating variable or stochastic spending patterns.

The growth factor g_{t+1} is determined by the asset allocation strategy being tested, which specifies the proportion of wealth invested in different asset classes. Specifically, we define asset allocation as a vector $\mathbf{a}_t$ at time t , where each element $a_{i,t}$ represents the proportion of wealth invested in asset class i . The portfolio growth factor is then calculated as:

$$g_{t+1} = 1 + \mathbf{a}_t^\top \mathbf{r}_{t+1} \quad (2)$$

Where $\mathbf{r}_{t+1}$ is a vector of returns for each asset class between time t and $t + 1$.

In a Monte Carlo simulation, we generate a large number of possible return paths. This means that for each time step, we have N different return vectors for our k asset classes. For each path, we simulate the investor’s wealth trajectory over time using the equations above. This means that equation 1 has the matrix form:

$$\mathbf{w}_{t+1} = \mathbf{w}_t \odot \mathbf{g}_{t+1} + \boldsymbol{\pi}_{t+1} \quad (3)$$

Where $\mathbf{w}_t$, $\mathbf{g}_{t+1}$, and $\boldsymbol{\pi}_{t+1}$ are vectors of length N representing the wealth, portfolio returns, and contributions or withdrawals for each of the N simulated paths at time t , and $\odot$ denotes element-wise (Hadamard) multiplication.

The growth factors $\mathbf{g}_{t+1}$ are determined by the asset allocation strategy and a $N \times k$ matrix of returns $\mathbf{R}_{t+1}$ for the k asset classes across the N simulated paths. The asset allocation strategy can be defined in various ways, the first of which is as a function of time (or age). In this case, we define the asset allocation at time t as $\mathbf{a}_t = f(t; \boldsymbol{\theta})$, where f is a function parameterised by a set of parameters $\boldsymbol{\theta}$.

This yields the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot f(t + 1; \boldsymbol{\theta}) \quad (4)$$

In this form, the model can accommodate a wide range of asset allocation strategies based on the investor’s age. This is similar to the approach that Anarkulova et al. (2025) use, and our first objective will be to replicate their findings for the optimal glide-paths, and to examine how these findings change when we modify the risk measures and spending assumptions to better reflect real-world investor behaviour, particularly in the New Zealand context.

Implications for Consilium and an Aside on Using Matrix Operations and Computational Efficiency

The effort taken to express the model in matrix form is not only for mathematical clarity; it also allows us to leverage vectorised operations through highly optimised linear algebra libraries (for example, `numpy` (Harris et al., 2020)), which can substantially accelerate simulation when path counts

and horizon lengths are large. In prior internal work on related Monte Carlo workflows, non-vectorised implementations relying heavily on for-loops have been used, which are computationally inefficient. By contrast, matrix operations allow array-level computation and can produce order-of-magnitude speed improvements. Applied in practice, this may materially improve analysis throughput.

The use of matrix operations will also make it easier when it comes to optimising the asset allocation strategy, as we can leverage automatic differentiation tools (e.g. `tensorflow` (Martín Abadi et al., 2015), `pytorch` (Paszke et al., 2019)) to compute gradients of the risk measure with respect to the parameters of the asset allocation strategy.

4.1 Modelling Dynamic Investor Behaviour

The model described above is flexible enough to accommodate a wide range of asset allocation strategies, but it also allows for the modelling of dynamic investor behaviour in terms of contributions and withdrawals. This is an important feature, as it allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor’s wealth.

4.1.1 Wealth-Responsive Spending Policy

Rather than having a fixed spending pattern, we can also model the investor’s spending as a function of their current wealth level. In some cases, retirees may have a fixed spending floor that they need to meet each year, regardless of the performance of their portfolio. In other cases, retirees may adjust their spending based on the performance of their portfolio, for example by reducing discretionary spending during periods of poor returns. To model this behaviour, we can define the net contribution (or withdrawal) at time t as a function of wealth: $\pi_t = g(W_t; \phi)$, where g is a function parameterised by a set of parameters ϕ . This allows us to simulate more realistic spending patterns that can vary over time and in response to changes in the investor’s wealth. For example, we could model a spending policy that has a fixed floor and a variable discretionary component that adjusts based on the investor’s current wealth level. This could be represented as:

$$\pi_t = -\max(S_{\text{floor}}, \alpha W_t) \quad (5)$$

Where S_{floor} is the fixed spending floor, and α is a parameter that determines how much of the investor’s wealth is allocated to discretionary spending.

Additionally, we can prevent the investor from withdrawing more than their current wealth by defining the net contribution as:

$$\pi_t = -\min(g_{t+1} W_t, \max(S_{\text{floor}}, \alpha W_t)) \quad (6)$$

Equally, we could model a spending policy that has a random component, for example by adding a stochastic term to the spending function modelling rare, but significant, unexpected expenses that retirees may face, such as healthcare costs or home repairs.

4.1.2 Wealth-Responsive Allocation Policy

Like with the spending policy, we can also model the asset allocation strategy as a function of both time and wealth. In the age-based glide-path framework, the hypothetical investor’s asset allocation strategy is only responsive to time (i.e. age), and does not take into account the investor’s current wealth level. In practice, an investor’s risk tolerance and spending needs may vary significantly depending on their accumulated wealth at any given time. For example, an investor with a large portfolio may be able to tolerate more risk than an investor with a smaller portfolio, even if they are the same age. To address this limitation, we will attempt to extend the framework using a method similar to that proposed by Mäkinen and Toivanen (2024), who introduce a control matrix that allows the asset allocation strategy to depend on both time and wealth.

In this framework, the asset allocation at time t and wealth W_t is given by $\mathbf{a}_t = C(t, W_t; \theta)$, where C is the control matrix parameterised by θ . This yields the portfolio growth factors as:

$$\mathbf{g}_{t+1} = 1 + \mathbf{R}_{t+1} \cdot C(t+1, W_t; \theta) \quad (7)$$

The way that this control-matrix works is that it defines a grid over the time and wealth dimensions, with each cell in the grid corresponding to a specific asset allocation. During the simulation, the investor’s current age and wealth level are used to determine which cell in the grid they fall into, and thus the corresponding allocation to the risky asset (the remainder of the portfolio is allocated to the safe asset).

This approach allows for a more nuanced asset allocation strategy that can adapt to the investor’s changing circumstances over time. For example, an investor who has accumulated significant wealth may be able to take on more risk, while an investor with lower wealth may need to adopt a more conservative approach. However, it also exponentially increases the number of parameters that need to be optimised, which can make any optimisation process more challenging.

However, we can reduce the search-space somewhat by using less granular grid points for the control matrix and interpolating between them to determine the asset allocation for time steps and wealth levels that fall between the defined grid points. This allows us to maintain a reasonable level of flexibility in the asset allocation strategy while keeping the number of parameters manageable. Moreover, it is unlikely that investors’ asset allocation policy will be meaningfully different for small increments in time or wealth, so using a coarser grid with interpolation is unlikely to significantly reduce the optimality of the resulting strategy, while making the optimisation process more tractable. This idea of interpolating between grid points was first proposed by Mäkinen and Toivanen (2024), who use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined wealth-grid points, however, we also propose that interpolation could be used for time steps that fall between the defined time-grid points as well, which would further reduce the number of parameters that need to be optimised.

To illustrate this more concretely, consider Figure 3, which shows a simple control matrix for a two-asset portfolio. The matrix has six time steps (0 to 5) and five wealth levels (0 to 4), giving a 6 by 5 grid. Each cell corresponds to a risky-asset allocation (with the remainder allocated to the safe asset). For instance, if an investor is at time step 2 with wealth level 3, the policy returns the allocation in that cell. For time or wealth values between grid points, we use the interpolation method proposed by Mäkinen and Toivanen (2024). For example, at time 2.5 and wealth 3.5, bilinear interpolation uses the four neighbouring grid points to determine allocation.

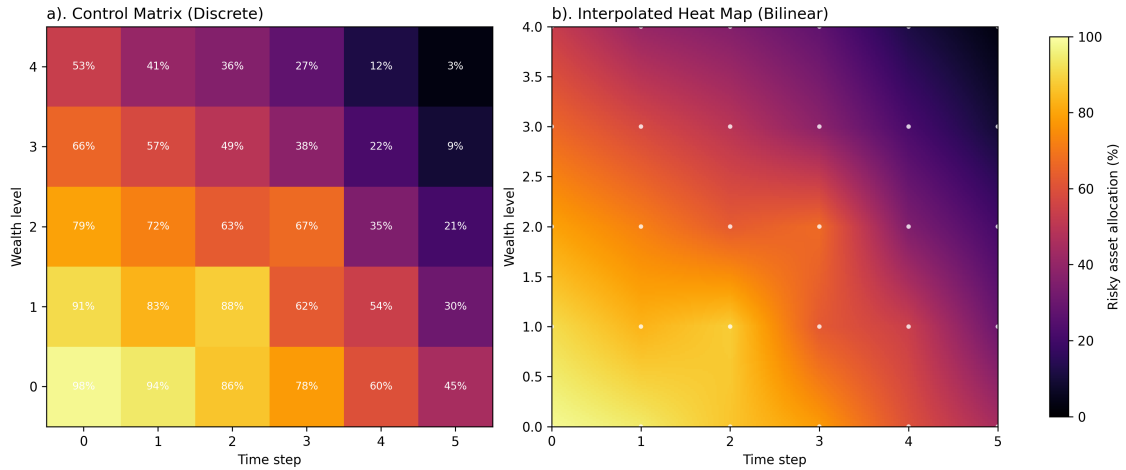


Figure 3: Illustrative example of a discrete control matrix (a) and bilinearly interpolated allocation surface (b).

For allocations with more than two asset classes, the control matrix can be extended to a higher-dimensional tensor, with each additional dimension representing an additional asset class. Mäkinen and Toivanen (2024) use bilinear interpolation to determine the asset allocation for wealth levels that fall between the defined grid points, ensuring a smooth transition between different allocation strategies.

4.2 Methods for Simulating Asset Returns

5 Methods for Optimising Asset Allocation Strategies

The key challenge in optimising an asset allocation strategy is that the parameter space can be extremely large, even for simple policy classes. For example, for a two-asset age-based glide-path with allocations in 10 percent increments over 60 years, there are 11 possible allocations at each time step, giving 11^{60} possible strategies. This search space is intractable, so numerical optimisation is required.

Merton (1969) use a formulation in which the investor maximises a lifetime utility function over consumption. The framework is mathematically tractable, but it relies on proportional consumption from wealth each period, which is often unrealistic for retirees facing fixed-dollar spending needs. In addition, the closed-form

solution assumes IID returns, precisely the return structure that Anarkulova et al. (2025) argue can bias conclusions about declining-risk optimality. For our purposes, this framework is therefore limited.

Anarkulova et al. (2025) use a recursive approach in which grid search is used to update allocation at each time step, with forward passes repeated until convergence. This approach is likely suitable for age-based glide-path optimisation, but extension to a time-by-wealth control matrix is less straightforward because the parameter count increases by roughly an additional order of magnitude.

In their paper, Mäkinen and Toivanen (2024) optimise a control-matrix allocation strategy using gradient-based methods and an adjoint formulation, which is computationally appealing for high-dimensional policy spaces. This provides a practical starting point for our own optimisation design, while still allowing us to adapt the objective function to better reflect retiree-relevant risk.

6 Optimal Lifetime Asset Allocation Strategies for New Zealand Retirees

Having established the simulation and optimisation framework, we now apply it to evaluate candidate lifetime allocation policies in the New Zealand retirement context.

6.1 Optimal Age-Based Glide-Paths

6.2 Optimal Wealth-Responsive Allocation Strategies

7 Discussion and Implications

8 Conclusion

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A Replication and Discussion of Control Matrix Optimisation as Proposed by Mäkinen and Toivanen (2024)

Mäkinen and Toivanen (2024) formulate their Monte Carlo optimisation in a way that is similar to the framework described in the main text. As in our setup, they use a control matrix that allows asset allocation to respond jointly to age and wealth. Their objective function is based on terminal wealth moments, specifically variance or semi-variance. They then compute gradients of this objective with respect to control-matrix parameters using adjoint methods, which enables efficient gradient-based optimisation (in their case, a quasi-Newton method with BFGS updates).

Because this method is mathematically elegant and computationally efficient, we have attempted to replicate it here. The code for this replication can be found in the accompanying repository, and the results are shown in Figure 4 for the mean-variance and mean-semi-variance cases.

In line with the setup described by Mäkinen and Toivanen (2024), the replication uses a Monte Carlo horizon of $T = 20$ years with time step $\Delta t = 0.25$, giving $N = 80$ time steps. The policy is represented on a time-wealth control grid with $M = 31$ wealth nodes, upper wealth bound $W_{\max} = 30$, and bilinear interpolation between nodes. In the notebook, wealth is clipped to $[0, W_{\max}]$ before interpolation. Optimisation is performed with L-BFGS-B (a quasi-Newton method) with box constraints.

For the one-risky-asset, no-leverage case used in Figure 4, the full parameterisation is summarised in Table 3.

Table 3: Parameter setup for the one-risky-asset mean-variance replication.

Parameter	Value
Risk-free rate (r)	0.03
Risky-asset drift (μ_1)	0.0795
Risky-asset volatility (σ_1)	0.15
Contribution rate (π)	0.1
Initial wealth (w_0)	1
Minimum risky allocation ($p_{\min}$)	0
Maximum risky allocation ($p_{\max}$)	1
Number of simulation paths (K)	10^5
Risk-aversion weight (λ)	0.1

The replication results are broadly consistent with those reported by Mäkinen and Toivanen (2024). However, they also highlight important limitations. First, as an objective for a retirement investor, terminal-wealth variance is not especially intuitive. In our replication, the terminal-wealth distribution becomes bimodal because the optimiser shifts to a highly conservative policy at high wealth levels, truncating the upper tail that would normally appear under a log-normal-like process. Although this behaviour is mathematically consistent with variance minimisation, it is not obviously aligned with investor welfare. A similar pattern appears under semi-variance, albeit at higher wealth levels. Because semi-variance is computed relative to the mean, higher-return strategies can increase the penalty by raising the mean itself, again encouraging upper-tail truncation. This reinforces the need for risk measures that are both interpretable and behaviourally relevant.

Additionally, the control matrix has limited effective resolution in early periods: many simulated paths map to only a few low-wealth nodes because initial wealth is far below the fixed upper wealth bound. This reduces the optimiser’s ability to distinguish trajectories precisely when policy differences likely matter most due to compounding. It also creates many redundant parameters at wealth levels that are never reached (upper-left region in Figure 4). For this reason, we propose calibrating the control-matrix upper bound to the simulated wealth distribution so that resolution is concentrated where paths actually lie.

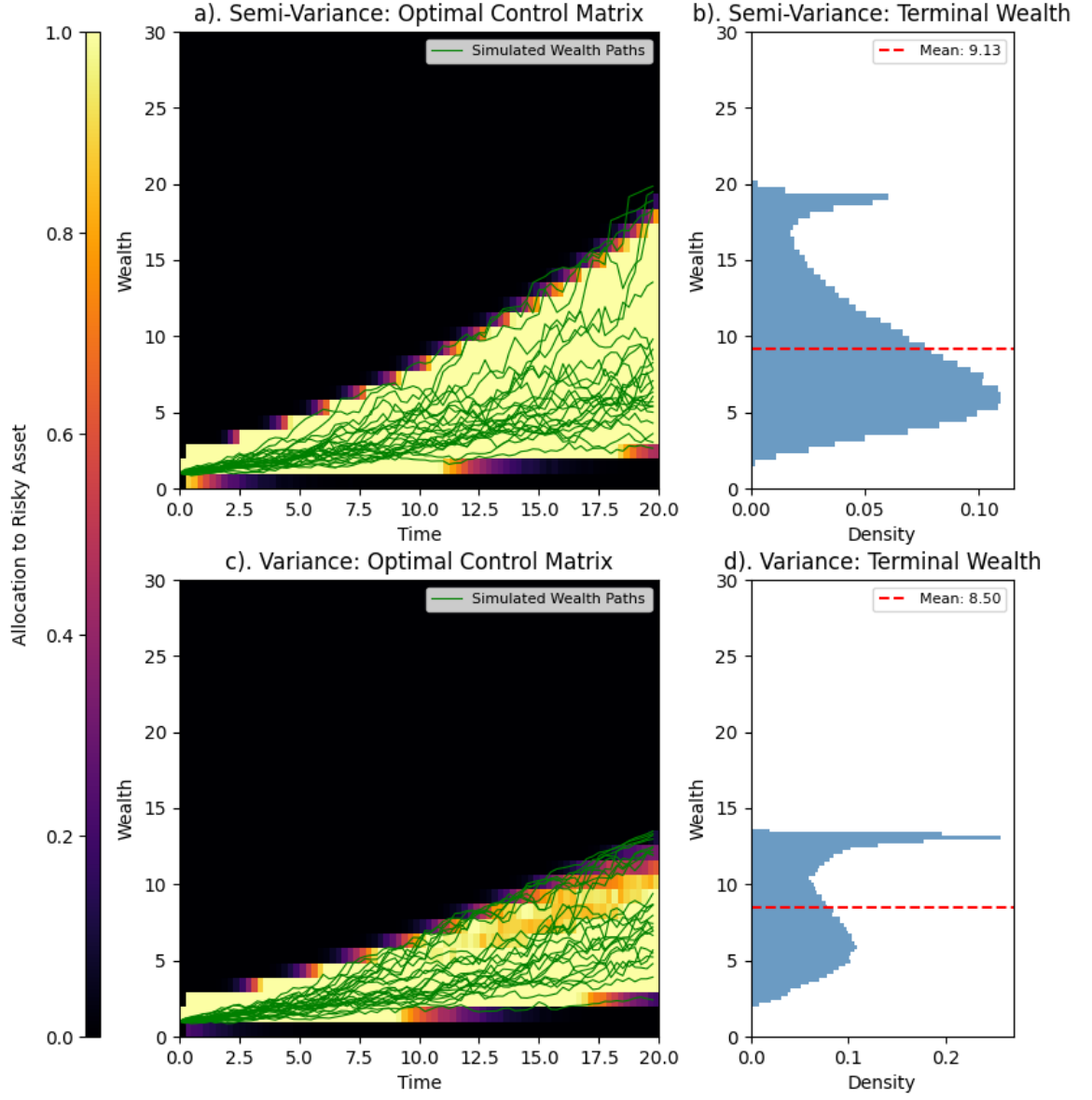


Figure 4: Replication of the control matrix optimization method proposed by Mäkinen and Toivanen (2024) for one risky asset and no leverage. The left column shows the optimal control matrix for the mean-semi-variance case (top) and the mean-variance case (bottom). The right column shows the corresponding histograms of terminal wealth for each risk measure respectively.